

Charles Galea



Homepage: <https://flask-portfolio-capgalea-charles-galeas-projects.vercel.app/>



Location: Melbourne, Victoria, Australia



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PROFESSIONAL SUMMARY

- **Organized, energetic and highly resourceful Data Scientist/Analyst** with 5 years of experience and prior 15+ year career as a biomedical researcher.
- **Experienced working with large complex datasets** to identify patterns and trends.
- Expertise in (1) **robotic process automation (RPA)**, (2) **dashboards and reports**, (3) programming languages **R, python, SQL**, (4) **process automation** and (5) **predictive modelling**.
- **Strong problem-solving skills** refined in industry and academia.
- **Works collaboratively with cross-functional teams** to develop strategies that meet business needs.
- **Excellent communication skills** conveying findings to technical and non-technical stakeholders.
- **Keen learner** who likes to apply state-of-the-art technology and current industry trends.
- Recipient of **5 awards at Ford** for projects leading to increases in efficiency and reducing cost.

AREAS OF EXPERTISE

Data Acquisition	Data Processing	Data Visualization	Statistical Analysis
Problem Solving	Process Automation (RPA)	Project Management	Agile
Effective Communication	Machine Learning	Strategic Thinking	

TOOLS

Python / R / SQL	Alteryx	Keras / PyTorch	Qlik Sense
Git	Google Cloud Platform	Jira	Hadoop
Jupyter Notebook	vsCode / RStudio	Selenium (Python)	LLMs

FORMAL EDUCATION

Master of Data Science, Royal Melbourne Institute of Technology (RMIT)
PhD, Biochemistry, University of Queensland

EMPLOYMENT HISTORY

Data Scientist

Feb 2020 – Jun 2024

Ford Motor Company

Ford of Australia is a subsidiary of the global Ford Motor Company that designs, engineers and imports cars, SUVs and trucks.

- Engaged with key stakeholders to identify data-based strategies to address business needs.

- End to end analytics: Developed timelines, identified datasets, created data pipelines for data mining / manipulation, modelling and statistical analysis.
- Facilitated feedback and impact assessments and implemented ongoing improvements.
- Held regular meetings with stakeholders and management to discuss progress and results.
- Prioritized workload based on importance and urgency.
- Performed statistical analysis and visualization of data to identify trends and patterns.
- Created dashboards and machine learning models (supervised and unsupervised).

Key Achievements

- Built **NLP/XG Boost ensemble machine learning model to categorize warranty claims** resulting in **30%** reduction in time spent by engineers on this task. *Ford Leadership Award (VCSE Innovation Awards 2020) and Ford Rewards and Recognition Program (Certificate of Appreciation).*
- Developed RPA **bot to extract purchase order information** resulting in identification of over **\$200 million in potential savings**. *Ford Rewards and Recognition Program (Certificate of Appreciation).*
- Developed RPA **bot for Ford leases to report issues with their vehicles**. Identified major quality issue and significant savings in warranty costs. *Ford Asia- Pacific Recognition Program award.*
- Developed **dashboard allowing engineers to monitor progress toward program milestone and gateways**. Dashboard rolled out to Ford teams globally.
- Created **dashboard highlighting key metrics and timeline toward resolving high-cost warranty issues**. Contributed toward team achieving **30%** reduction in warranty costs for year 2020. *Ford Asia-Pacific Recognition Program award.*

Data Science Intern

Jul 2019 – Oct 2019

Ford Motor Company

- Processed, performed static analyses and visualization of dataset to identify trends and patterns.
- Trained and optimized various machine learning models.
- Report highlighting results presented the report to management.

Key Achievements

- Developed a regression random forest **machine learning model to predict prototype part weights** based on similar parts previously designed by the Ford team and other automotive companies which predicted prototype part weights to within 5-10% of actual value.

Senior Postdoctoral Fellow (Drug Development)

Dec 2010 – Apr 2019

Monash Institute of Pharmaceutical Sciences (MIPS)

- Genetic and functional studies of several neurodegenerative diseases (collaboration with clinicians and researchers at Murdoch Children's Research Institute). Development of (1) drugs for the treatment of several autoimmune diseases and (2) novel antibiotics targeting antibiotic-resistant bacteria.

Key Accomplishments

- *Published 14 articles* in peer-reviewed scientific journals and a **book chapter**.

Senior Research Fellow (Drug Development)

Feb 2008 – Dec 2010

Walter and Eliza Hall Institute of Medical Research

- Development of novel therapeutics based on cone snail toxins for the treatment of Multiple Sclerosis.

Key Accomplishments

- *Published 3 papers in peer-reviewed scientific journals.*

FURTHER EDUCATION

Participant, AI Pathfinder Program (AI Startup course for founders)

SMEC AI (The AI Pathfinder program is powered by SMEC AI, which is funded by the Department of Industry, Science and Resources through the AI Adopt program.)

Deep Learning Specialization

Coursera F93WV5XKDD2D

Google Cloud Big Data and Machine Learning Fundamentals

Coursera PANX6DRATJ45

Statistical Learning

Stanford Online OpenEdx 6485c12d55b54a45ad0ee3f31336f519

Develop GenAI Apps with Gemini and Streamlit Skill Badge

Google

Intro to Snowflake for Devs, Data Scientists, Data Engineers

Coursera KUMDKCKQ08VS

